

SKILL_WEEK-07

Task 1: Process Synchronization and Child Process Monitoring Using waitpid()

```
UW PICO 5.09
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("Parent Process PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process PID: %d\n", getpid());
        printf("Child is performing its task...\n");

        sleep(3);

        printf("Child task completed.\n");

        exit(5);
    }
    else
    {
        printf("Parent created Child PID: %d\n", pid);
        printf("Parent is waiting for the child using waitpid()...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status))
        {
            printf("Child terminated normally.\n");
            printf("Child Exit Status: %d\n", WEXITSTATUS(status));
        }
        else
        {
            printf("Child terminated abnormally.\n");
        }

        printf("Parent process resumes execution.\n");
    }

    return 0;
}
```

Fig 1: C program demonstrating process synchronization and child monitoring using waitpid()

```
[amrutha@Amruthas-MacBook-Air OSSP % gcc src/skill17_task1.c -o bin/skill17_task1
[amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill17_task1
Parent Process PID: 66689
Parent created Child PID: 66690
Parent is waiting for the child using waitpid()...
Child Process PID: 66690
Child is performing its task...
Child task completed.
Child terminated normally.
Child Exit Status: 5
Parent process resumes execution.
amrutha@Amruthas-MacBook-Air OSSP %
```

Fig 2: Execution output showing parent-child synchronization and child exit status using waitpid()

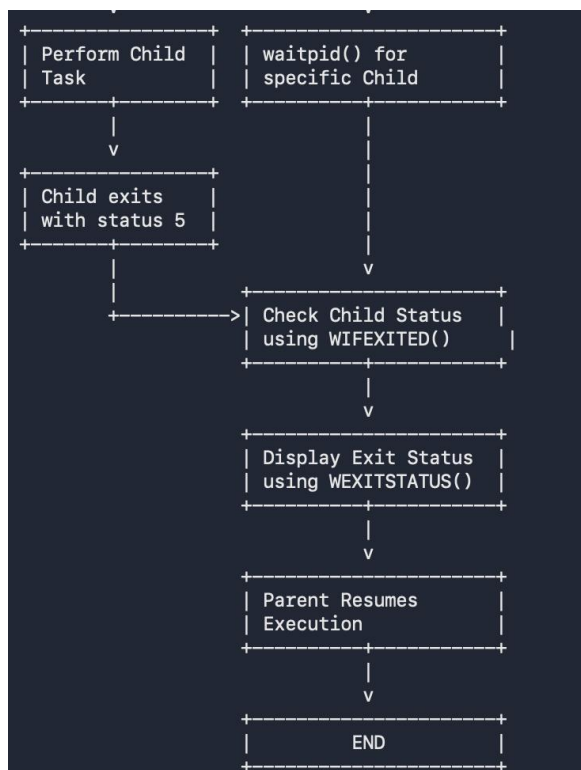
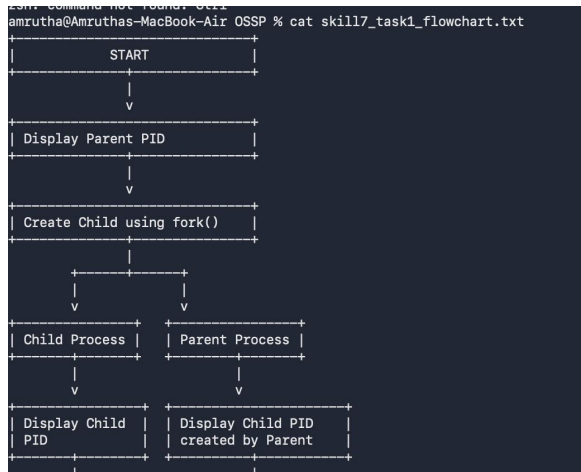
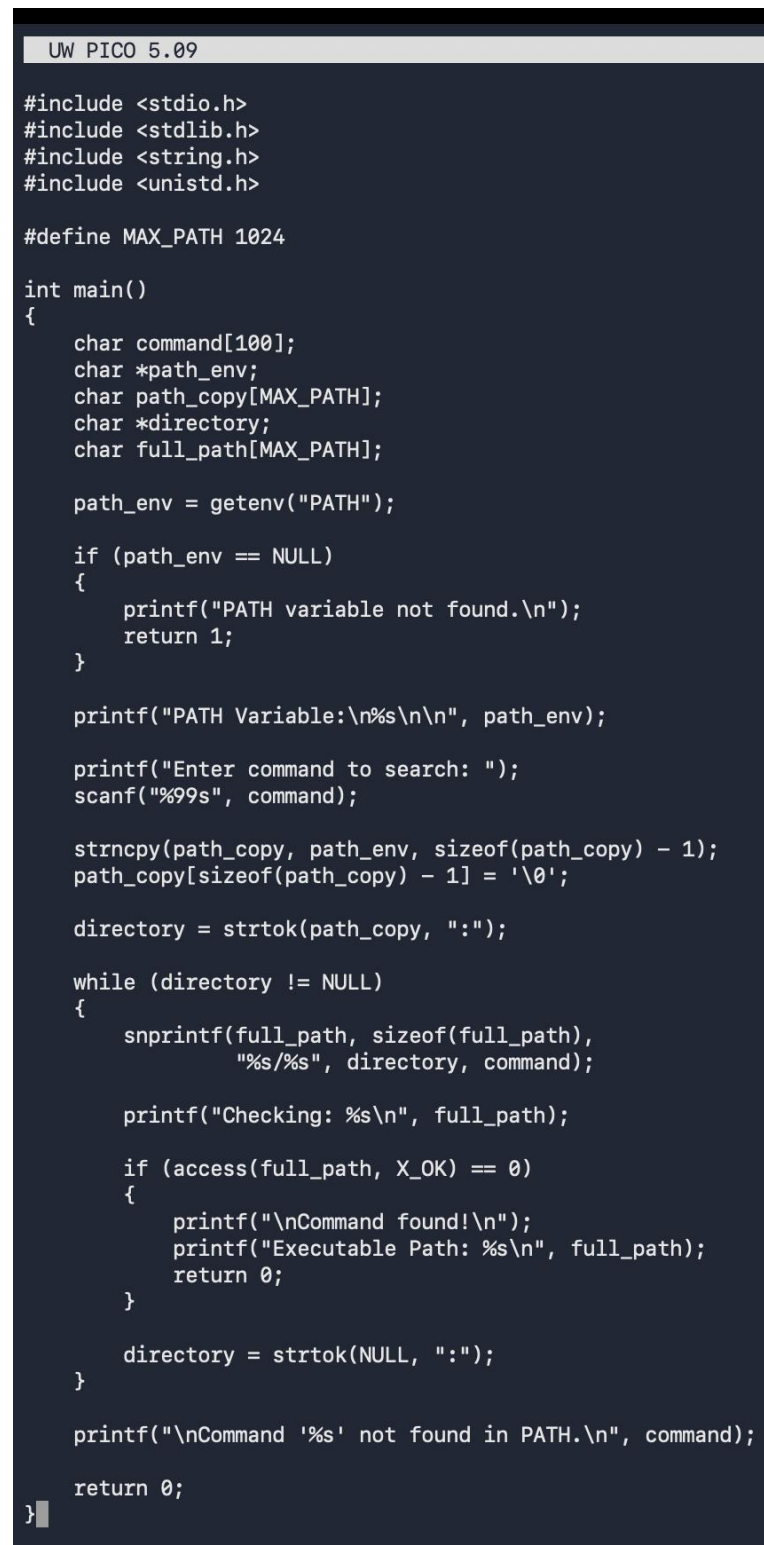


Fig 3: Flowchart for process synchronization and child monitoring using waitpid()

Result:

The child process was created successfully using fork() and monitored using waitpid(). The parent process waited for the specific child to complete, retrieved its exit status, and resumed execution after successful synchronization.

Task 2: Retrieve the PATH variable, parse search directories, locate executables, verify permissions, handle missing commands, and test command resolution.



```
UW PICO 5.09

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX_PATH 1024

int main()
{
    char command[100];
    char *path_env;
    char path_copy[MAX_PATH];
    char *directory;
    char full_path[MAX_PATH];

    path_env = getenv("PATH");

    if (path_env == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH Variable:\n%s\n\n", path_env);

    printf("Enter command to search: ");
    scanf("%99s", command);

    strncpy(path_copy, path_env, sizeof(path_copy) - 1);
    path_copy[sizeof(path_copy) - 1] = '\0';

    directory = strtok(path_copy, ":");

    while (directory != NULL)
    {
        snprintf(full_path, sizeof(full_path),
                 "%s/%s", directory, command);

        printf("Checking: %s\n", full_path);

        if (access(full_path, X_OK) == 0)
        {
            printf("\nCommand found!\n");
            printf("Executable Path: %s\n", full_path);
            return 0;
        }

        directory = strtok(NULL, ":");
    }

    printf("\nCommand '%s' not found in PATH.\n", command);

    return 0;
}
```

Fig 4: C program for retrieving the PATH variable and resolving executable commands

Test Case - 1:

```
amrutha@Amruthas-MacBook-Air OSSP % gcc src/skill17_task2.c -o bin/skill17_task2
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill17_task2
PATH Variable:
/opt/homebrew/opt/openjdk@21/bin:/opt/homebrew/opt/postgresql@17/bin:/opt/homebrew/bin:/opt/homebrew/sbin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin:/opt/homebrew/bin
Enter command to search: ls
Checking: /opt/homebrew/opt/openjdk@21/bin/ls
Checking: /opt/homebrew/opt/postgresql@17/bin/ls
Checking: /opt/homebrew/bin/ls
Checking: /opt/homebrew/sbin/ls
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/ls
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/ls
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/ls
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/ls
Checking: /usr/local/bin/ls
Checking: /System/Cryptexes/App/usr/bin/ls
Checking: /usr/bin/ls
Checking: /bin/ls
Command found!
Executable Path: /bin/ls
```

Fig 5: Successful resolution of the ls command using the PATH environment variable

Test case - 2:

```
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill17_task2
PATH Variable:
/opt/homebrew/opt/openjdk@21/bin:/opt/homebrew/opt/postgresql@17/bin:/opt/homebrew/bin:/opt/homebrew/sbin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin:/opt/homebrew/bin
Enter command to search: gcc
Checking: /opt/homebrew/opt/openjdk@21/bin/gcc
Checking: /opt/homebrew/opt/postgresql@17/bin/gcc
Checking: /opt/homebrew/bin/gcc
Checking: /opt/homebrew/sbin/gcc
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/gcc
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/gcc
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/gcc
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/gcc
Checking: /usr/local/bin/gcc
Checking: /System/Cryptexes/App/usr/bin/gcc
Checking: /usr/bin/gcc
Command found!
Executable Path: /usr/bin/gcc
```

Fig 6: Resolving the compiler executable using PATH directory search

Test case - 3:

```
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill17_task2
PATH Variable:
/opt/homebrew/opt/openjdk@21/bin:/opt/homebrew/opt/postgresql@17/bin:/opt/homebrew/bin:/opt/homebrew/sbin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin:/opt/homebrew/bin
Enter command to search: abcxyzcommand
Checking: /opt/homebrew/opt/openjdk@21/bin/abcxyzcommand
Checking: /opt/homebrew/opt/postgresql@17/bin/abcxyzcommand
Checking: /opt/homebrew/bin/abcxyzcommand
Checking: /opt/homebrew/sbin/abcxyzcommand
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/abcxyzcommand
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/abcxyzcommand
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/abcxyzcommand
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/abcxyzcommand
Checking: /usr/local/bin/abcxyzcommand
Checking: /System/Cryptexes/App/usr/bin/abcxyzcommand
Checking: /usr/bin/abcxyzcommand
Checking: /bin/abcxyzcommand
Checking: /usr/sbin/abcxyzcommand
Checking: /sbin/abcxyzcommand
Checking: /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin/abcxyzcommand
Checking: /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin/abcxyzcommand
Checking: /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin/abcxyzcommand
Checking: /pkg/env/global/bin/abcxyzcommand
Checking: /opt/homebrew/bin/abcxyzcommand
Command 'abcxyzcommand' not found in PATH.
```

Fig 7: Handling a command that is not present in the PATH directories

```

amrutha@Amruthas-MacBook-Air OSSP % nano skill7_task2_flowchart.txt
amrutha@Amruthas-MacBook-Air OSSP % cat skill7_task2_flowchart.txt
+-----+
| START |
+-----+
  |
  v
+-----+
| Retrieve PATH using getenv() |
+-----+
  |
  v
+-----+
| Read Command Name |
+-----+
  |
  v
+-----+
| Split PATH using ':' |
+-----+
  |
  v
+-----+
| Build Full Path |
| directory/command |
+-----+
  |
  v
+-----+
| Check Execute Permission |
| using access(..., X_OK) |
+-----+

```

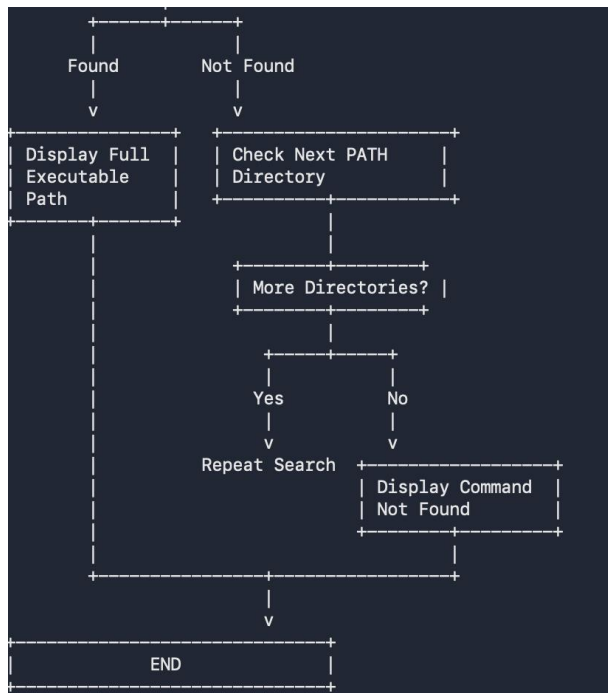


Fig 8: Flowchart for PATH variable parsing and executable command resolution

Result:

The PATH environment variable was retrieved and parsed successfully. Executable commands were located by searching PATH directories and checking execution permission using `access()`. Missing commands were also handled correctly.